Arista Access Node KOR-ACC-66 Failure Recovery SOP for ALARM_9007

Node: KOR-ACC-66 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9007

Description

The failure domain of ASIC memory parity errors, line card crashes, and power supply redundancy loss in the Arista Access Node KOR-ACC-66 at the Koramangala Datacenter is a critical failure scenario. This is due to the inherent design of the Arista EOS (Extensible Operating System) that relies on the ASIC's ability to manage and process vast amounts of data. In the event of a hardware or environmental fault, the ASIC's memory parity errors can lead to a cascade failure, ultimately causing the line card to crash. Furthermore, the loss of power supply redundancy can exacerbate this situation, rendering the node inoperable. The Arista Access Node's role as a critical component in the network infrastructure necessitates a swift and accurate response to this failure scenario, minimizing downtime and ensuring network resilience.

Pre-requisite Checks

- Check: Verify that the Koramangala Datacenter is accessible and that remote hands are dispatched to the site.

- Check: Inspect the node's redundant links to ensure that they are functioning properly and can be used as a failover path.

- Check: Review the node's environmental logs to determine if there have been any recent power supply fluctuations or other environmental issues that may have contributed to the failure.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes due to hardware failure	Medium
Loss of power supply redundancy	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource usage, Expected Output: System resource usage: [ASIC memory usage percentage], Description: This command provides insight into the ASIC's memory usage, helping to identify potential memory parity errors.

Command: show interface [interface ID] stats, Expected Output: Interface [interface ID] statistics: [error count, discard count], Description: This command provides detailed statistics on the interface, helping to identify potential issues with the line card.
```

```
Command: show power supply redundancy, Expected Output: Power supply redundancy status: [online/ offline], Description: This command provides information on the power supply redundancy status, helping to identify potential issues with the power supply.
```

Resolution Steps

Step 1: Step: Replace the failed ASIC, Command: replace system asic, Description: Replacing the failed ASIC is the most critical step in resolving this failure scenario. The Arista EOS will automatically recognize the new ASIC and configure it accordingly.

Step 2: Step: Replace the failed line card, Command: replace system line-card [line-card ID], Description: Replacing the failed line card is essential to restore network connectivity. The Arista EOS will automatically recognize the new line card and configure it accordingly.

Step 3: Step: Restore power supply redundancy, Command: enable power supply redundancy, Description: Restoring power supply redundancy is crucial to ensure that the node remains operational in the event of another failure.

Post-Check Verification

```
Command: show system resource usage, Expected Output: System resource usage: [ASIC memory usage percentage <= 50%], Description: This command verifies that the ASIC memory usage is within a safe range, indicating that the replacement was successful.

Command: show interface [interface ID] stats, Expected Output: Interface [interface ID] statistics: [error count, discard count = 0], Description: This command verifies that the interface is functioning correctly, indicating that the replacement was successful.

Command: show power supply redundancy, Expected Output: Power supply redundancy status: [online], Description: This command verifies that the power supply redundancy is restored, indicating that the replacement was successful.
```

Roll-back Procedures

- Step: Restore the original ASIC, Command: replace system asic original, Description: Restoring the original ASIC is necessary in case the replacement caused any issues.

- Step: Restore the original line card, Command: replace system line-card [line-card ID] original, Description: Restoring the original line card is essential to maintain network connectivity.

- Step: Disable power supply redundancy, Command: disable power supply redundancy, Description: Disabling power supply redundancy is necessary to prevent any potential issues with the power supply.